

DAY 9 / 30 · AZURE DATA ENGINEER SERIES

Blob Storage Tiers & Lifecycle

Hot, Cool, Archive & automated lifecycle rules — explained with real-time examples, tips & interview questions

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1 Hot Tier

Designed for data accessed frequently — highest storage cost, but lowest cost to read/access.

◆ REAL-TIME EXAMPLE

An e-commerce site stores today's and this week's order data in Hot tier because dashboards and apps query it constantly throughout the day.

💡 TIP

Say "Hot = frequent access, higher storage cost, lower access cost" as one clean sentence — interviewers like concise tradeoff statements.

2

Cool & Cold Tier

Designed for data accessed infrequently but still needs quick access when needed — lower storage cost than Hot, but higher cost per access.

◆ REAL-TIME EXAMPLE

Last quarter's sales reports are moved to Cool tier — still needed occasionally for audits or year-over-year comparisons, but not queried daily.

💡 TIP

Mention the minimum storage duration (30 days for Cool) — forgetting this in interviews is a common small mistake that's easy to avoid.

3 Archive Tier

Designed for data rarely accessed — cheapest storage cost, but data must be "rehydrated" (which takes hours) before it can be read.

◆ REAL-TIME EXAMPLE

A healthcare company keeps 7-year-old patient records in Archive tier to meet compliance requirements — extremely cheap to store, and rarely, if ever, accessed.

💡 TIP

Mention "rehydration" specifically — it's the exact word interviewers expect, and shows you understand Archive isn't instantly readable like Hot or Cool.

4

Lifecycle Management Policies

Rules that automatically move data between tiers (or delete it) based on age, without any manual intervention.

◆ REAL-TIME EXAMPLE

A lifecycle policy automatically moves log files to Cool tier after 30 days, then to Archive after 90 days, then deletes them after 365 days — all configured once, running forever.

💡 TIP

Say "set it once, forget it" — lifecycle policies save significant manual effort and cost, and interviewers want to hear you value automation.

Choosing the Right Tier — Cost Thinking

The right tier isn't about the data type — it's about access frequency. The same file type can belong in different tiers depending on how often it's read.

◆ REAL-TIME EXAMPLE

Two identical CSV files — one is this month's active sales data (Hot), the other is a 3-year-old backup (Archive) — same format, completely different tier based on usage pattern, not content.

💡 TIP

If asked "which tier would you use," always ask back (or state the assumption): "That depends on how often it's accessed" — this shows real cost-conscious engineering thinking.

Bonus Interview Questions

- ? What is rehydration, and how long does it typically take from Archive tier?
- ? What happens if you delete a Cool tier blob before its 30-day minimum duration?
- ? How would you design a lifecycle policy for application log files?
- ? Can lifecycle policies be applied to specific folders only, or the whole container?
- ? What's the tradeoff between storage cost and access cost across tiers?

🎯 Communication Tips — Practice Daily

Tier questions are cost-optimization questions in disguise — think and speak like an engineer managing a budget.

- 1 **Always frame answers around access frequency**, not file type — this is the core decision factor interviewers want to hear.
- 2 **Mention the cost tradeoff explicitly**: "cheaper storage, more expensive access" — say it as one clear sentence.
- 3 **Bring up lifecycle policies proactively** — it shows you think about automation, not just manual tier selection.
- 4 **Use the word "rehydration"** when discussing Archive — it's precise and shows real familiarity.
- 5 **Give a real numbers-based example** if you can (e.g., "30 days," "90 days") — specifics beat vague answers.

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